Unit 2 Assignment 1: Linear Algebra & NumPy Practice

Work Type: Group Assignment

Task Scenario

Each group will create a small dataset as a matrix.

Example: student scores, customer data, product prices, image pixels, or user features.

```
data = np.array([
    [80, 90, 85],
    [70, 75, 78],
    [95, 88, 92],
    [60, 65, 70]
])
```

Part 1: NumPy & Math Foundations

Students must apply:

1. Define examples of:
    - Scalar
    - Vector
    - Matrix
    - Tensor
2. Create NumPy arrays for each one.
3. Show the shape and dimensions.
4. Explain the difference between:
    - (n,)
    - (n,1)
5. Apply element-wise operations:
    - Addition

    - Subtraction
    - Multiplication
    - Division
6. Apply aggregations:
    - Sum
    - Mean
    - Min
    - Max
    - Standard deviation

Part 2: Vectors & Linear Algebra

Students must apply:

1. Create two vectors v and w.
2. Compute:
    - Vector addition
    - Scalar multiplication
    - Dot product
    - Norm
    - Cosine similarity
3. Create row vector and column vector.
4. Explain the meaning of dot product:
    - positive
    - zero
    - negative
5. Create two vectors and test:
    - Are they linearly independent?

    - Do they form a basis for R^2?
6. Create a linear combination:

a * v + b * w

7. Apply a linear transformation using a matrix:
    - Scaling
    - Rotation or shear

Final Question

Each group answers:

Why are vectors, matrices, and transformations important in Machine Learning?

Deliverables

- Jupyter Notebook .ipynb
- Math explanation + NumPy code
- Output for every task
- Short group presentation: 3–5 minutes